

IE48L HCID
Assignment 2: HIERARCHICAL TASK ANALYSIS (HTA)
(Independent work is required)
Due date: 25 March 2026

Hierarchical Task Analysis (HTA) is the most popular ergonomics method that can help to understand and discover the structure of a task, system or behaviour. **The overall aim of HTA is to identify actual or possible sources of performance failure and to propose suitable remedies**, which may include **modifying the task design** and/or **providing appropriate training**. The process of HTA involves systematically decomposing tasks into subtasks to any desired level of detail. These are grouped as **plans** that specify how the tasks might be performed in practice.

The assignment contains three parts:

- (1) Conduct a hierarchical task analysis (HTA) on a user's mental model of **sharing photos via email using the phone**.
- (2) Then, **create another HTA**, this time on your own, **for an alternative way of doing the same action**.
- (3) Lastly, you will **compare the two models** and elaborate on the **pros and cons of the two ways to approach the task**.

We covered HTAs in class, and at the end of the page, you will also find additional sources on HTAs.

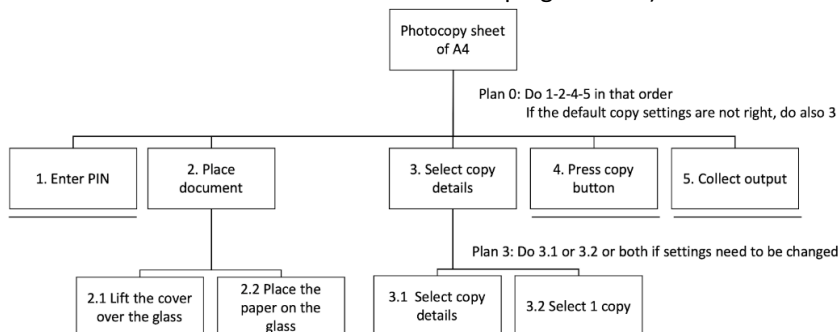
Prepare for the tasks:

1. Read some of the resources provided (refer to IE Moodle) on HTAs and revisit course slides so you are more familiar with its principles and structure.
2. Familiarize yourself with how to share a photo via email on a mobile phone.

Task 1: Carry out an HTA based on a user's mental model of sharing photos via email

Find a friend and ask them to show you how to share photos from a phone via email. Let the user explain the steps they take. Make notes while you observe the user. Do not seek to correct the user if the sequence of actions has inefficiencies. If the user does not know how to perform this task, continue until you understand what the user is trying to do, and then show how the task can be performed.

Once you think you know what the user's task structure is like, thank the user and **prepare an HTA diagram that describes how the user went about sharing photos on their phone using email**. (Examples of such HTA diagrams can be found in lecture notes (refer to the course website. You can use them as the main reference when developing the HTA).



Note: It may be easier to first use a textual representation of the task (see lecture slides for "Non-graphical HTA") and then sketch the diagram with pen and paper before preparing the final diagram. These methods may make it easier for you to create the final diagram.

Task 2: Create an alternative HTA diagram for sharing photos via email using a phone

Your participant has probably demonstrated photo sharing in either of the following two general ways:

1. Go to Photo Gallery, choose Share, select Email.
2. Go to Email, choose Add attachment, choose the Photo...

In this Task 2, you have **to create an alternative HTA diagram**, depending on which of the two methods listed above was the one that your participant showed to you. Select the sharing method that your participant did not demonstrate, and perform a HTA for that sharing method on your own, and draw an HTA diagram.

(The participant may have of course demonstrated photo sharing using a third alternative method too. In that case you can choose either of the two methods listed above.)

Task 3: Comparison of the two HTA diagrams

Compare the two diagrams and identify the similarities and differences. Analyse the pros and cons of the two task structures by their execution speed, vulnerability to mistakes, and simplicity.

How to return this assignment

Your submission should include:

- 1) HTA diagram from Task 1 and
- 2) HTA diagram from Task 2, and
- 3) a comparison of the two.

This assignment is carried out independently.

Notes:

1. Refer to the provided resources (e.g., Moodle) for further information.
2. You may contact your friends through WhatsApp, Skype, or Zoom to complete the assignment.